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UGANDA NATIONAL EXAMINATIONS BOARD
UGANDA ADVANCED CERTIFICATE OF EDUCATION
NOVEMBER - DECEMBER, 2023

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MITYANA STANDARD SECONDARY SCHOOL
S.5 PHYSICS ONE
END OF TERM TWO 2025
TIME: ~~Answer any two~~ 1hr: 50 min
INSTRUCTIONS: Answer All.
Assume $g = 9.8 ms^{-2}$

ITEM ONE

Consider a viscous fluid undergoing a steady flow through a pipe of circular cross sectional area. Since there is a viscous drag, velocity which varies from maximum at the centre of the pipe to zero at the walls makes rate of volume flow $\left(\frac{V}{t}\right)$ depend on:

(i) Coefficient of viscosity of the liquid.
(ii) Radius r of the pipe.
(iii) Pressure gradient $\left(\frac{P}{L}\right)$ where P is the pressure difference between the ends of the pipe and L is the length of the pipe.

Task:

Using the knowledge of Physics:

a) Derive an expression to derive an expression for the volume flow rate of a liquid passing any section of the pipe in ~~time~~ time, t.

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Hint: Assume coefficient of viscosity

$$\eta = ML^{-1}T^{-1}$$

Item two:

In an attempt to ~~determine~~ carry out some scientific investigation, the following results were based on as summarised in the table below:

Voltage, V(V)	Current I(A)	mass collected in 300s/g
7.4	2.6	5.8
10.0	3.6	11.3

Task

As a student of physics and using the knowledge of physics:

a) State and explain which scientific investigation was being conducted.

b) Briefly out line the steps involved in the determination of the investigation above.

c) Compute the value of the scientific investigation from the above results and determine rate of energy lost to the nearby surrounding.

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Item three

In an attempt to know about the various weather conditions and temperatures of different places and substances, some instruments say thermometers are used to measure temperatures of both small and big bodies.

Task:

Use your knowledge of Physics to:

a) Define thermometric property and its characteristics
b) State and justify the type of thermometer that uses pressure to perform its activities. Relevant diagram, procedures and advantages are required.
c) Describe how a scale of temperature can be established basing on the above thermometer.

— END —

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